

What Gives You Away?

How Linguistic Cues Move User Readouts Across Three Language Models

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August 2026

Abstract

What makes a language model read one user as older, more educated, wealthier, or a different gender? We place controlled linguistic edits inside matched multi-turn conversations and measure their effect on four TalkTuner-style user readouts. The main experiment evaluates 1,025 pairs over 100 independent base conversations in Llama 3.2 3B. A fixed 505-pair panel is also evaluated in Llama 3.1 8B and OLMo 2 32B. Each model has its own five-seed ensemble of full-conversation probes. Every evaluation pair differs at one user turn; surrounding user and assistant messages are fixed.

In the full 3B experiment, 30 of 36 cue-by-readout cells survive correction; 28 move more than a matched lexical-edit control. Disclosure-to-age, grammar-to-education, orthography-to-education, price-to-socioeconomic-status, and slang-to-age form the clearest pattern. On the shared panel, the 8B and 32B maps correlate .927 and .849 with 3B, retain the direction of all 30 corrected 3B effects, and keep 29 corrected-significant.

Cue ordering is robust to independently worded conversation shells and readout-phrase paraphrases. Persistence is cue-dependent: direct disclosure remains readable after later turns, while several style effects attenuate. These are causal changes in fixed model readouts, not verified facts about a person. The results identify a stable core while showing that weak effects and individual cue-to-attribute assignments should not be universalized.

1 Introduction

A user asks for an affordable hotel. Another makes the same request but says that price is no concern. One writes in fragments; another uses a subordinate clause. A third adds an emoji. These differences may be peripheral to the task, yet they can change the person a conversational model appears to be answering.

Writing has long been known to carry information about its author (Stamatatos, 2009; Schler et al., 2006). In a dialogue system, however, the relevant question is not simply whether demographic information can be predicted from a corpus. It is whether a small, controlled change to one user turn alters the model’s internal reading of that user while the rest of the conversation remains unchanged.

We study that question with the age, gender, education, and socioeconomic readouts used by TalkTuner (Chen et al., 2024). A linear probe converts a hidden state into class logits, giving a repeatable instrument for measuring how an edit changes the model’s elicited impression. We fit a separate five-seed probe ensemble for each of three models. Rows sharing a source generator and numeric index remain together during training and evaluation, so related source rows cannot cross partitions.

The experimental context matters. The probes were trained on full conversations, so every primary measurement is made in a full conversation as well. Two independently worded neutral preludes place the edited message at the current user turn. Two further conditions move that message one or two user turns into the past. The paired conversations always share the same assistant messages, preventing a generated response from becoming an uncontrolled mediator.

The study makes three contributions. First, it supplies a context-matched map from ten controlled edit categories to four user readouts. Second, it tests that map across 3B, 8B, and 32B checkpoints with independently trained probes and model-comparable endpoints. Third, it tests changes in conversation wording, probe seed, and readout phrase and measures how selected effects decay as later turns enter the context.

The central result is not that every surface feature matters, or that all models read cues identically. It is that a stable core of the cue map survives substantial changes in model size and family, while some assignments—most clearly emoji-to-gender—do not. What a model reads from a user is a cue-specific, temporally updated impression rather than a universal profile.

2 Related work

User representations. TalkTuner trains linear probes for a conversational user’s age, gender, education, and socioeconomic status, then exposes those estimates in an interactive dashboard (Chen et al., 2024). Related work examines implicit personalization and stereotype-driven user representations (Neplenbroek et al., 2025). We focus on the input evidence that moves a fixed user-reading interface.

Probing and interpretation. The probing literature distinguishes decodable information from information that a model necessarily uses (Hewitt and Liang, 2019; Voita and Titov, 2020; Elazar et al., 2021). Our paired interventions support a narrow causal claim: the edited cue changes the probe’s readout of a fixed model state. They do not show that the probe label is true of the user or that the same direction governs the model’s generated response.

Prompt sensitivity is a related measurement problem: small changes to an evaluation prompt can alter model scores without changing the underlying task (Hua et al., 2025). We test this directly by repeating the cue matrix under three paraphrases of each attribute’s readout phrase.

Style and demographic inference. Stylometric systems use vocabulary, syntax, spelling, and punctuation to infer properties of authors and documents (Stamatatos, 2009). Such correlations can reflect both genuine population structure and dataset-specific stereotypes. A paired design asks a different question: after holding the request and conversation fixed, which individual cue changes which model reading?

3 Method

3.1 Model and readouts

The reference model is Llama 3.2 3B Instruct. We repeat the study on Llama 3.1 8B Instruct, a larger checkpoint from the same model family, and OLMo 2 32B Instruct, a larger independently developed model (Grattafiori et al., 2024; OLMo Team et al., 2024). Immutable revisions are recorded in the repository. This is a cross-model comparison, not a factorial test of parameter count: model size, family, and training recipe are not independently varied.

Each conversation is formatted with the relevant model’s chat template. At the start of a new assistant turn, we append an attribute-specific phrase such as “I think the education level of this user is”. A multinomial logistic regression reads the hidden state at the final token of that phrase. For hidden state h ,

$$p(y = c \mid h) = \text{softmax}_c(Wh + b).$$

The four readouts are age, gender, education, and socioeconomic status. We fit the probes separately in each model using the same complete TalkTuner conversations and recipe. For each attribute, 300 label-complete sets sharing a generator folder and numeric source index are sampled. Each set stays within one 70/15/15 train, layer-selection, or test partition. The middle split selects the layer; the final split is used only for reporting. We fit five split seeds. Table 1 reports held-out balanced accuracy. Every attribute and seed exceeds the pre-specified .75 gate.

Readout	Llama 3.2 3B	Llama 3.1 8B	OLMo 2 32B
Age	.963	.958	.972
Gender	.929	.949	.889
Education	.957	.957	.966
Socioeconomic status	.975	.970	.978

Table 1: Mean held-out balanced accuracy across five independently split probe seeds. The lowest individual score is .822 for the 32B gender readout.

For a matched edit from conversation A to conversation B , the class-logit change is

$$\Delta z_c = w_c^\top (h_B - h_A).$$

The intercept cancels. Both variants use the same model, conversation shell, readout phrase, probe seed, and readout position. The outcome for one probe is $\max_c |\Delta z_c|$. This measures how far the edit moves the attribute readout without privileging a particular class direction. Because every model has a separately fitted probe, raw logit magnitudes are meaningful only within a model; cross-model analyses compare ranks, signs, corrected significance, dominant readouts, and dimensionless retention ratios.

3.2 Controlled pairs

The dataset contains 1,025 pairs over 100 user messages about travel, cooking, small talk, technical support, health, finance and shopping, and work or study. Each base contributes controlled cue edits and a neutral synonym control. Table 2 summarizes the complete dataset.

Category	Pairs	Category	Pairs
Orthography	184	Explicit disclosure	111
Emoji	100	Grammatical complexity	100
Random-synonym control	100	Contraction/formality	94
Slang	94	Affect language	82
US-to-UK spelling	82	Price language	78

Table 2: Released controlled pairs. Orthography includes lowercasing, punctuation removal, and typographical errors. Random synonym replacements provide a matched lexical-movement floor.

A price edit may add a tight-budget or price-is-no-concern qualifier. An emoji edit appends one symbol. A grammar edit rewrites one sentence with more complex syntax while preserving the request. Orthographic edits alter capitalization, punctuation, or spelling. Pairs were reviewed for naturalness, a clear single target manipulation, and preservation of the underlying request before model readouts were inspected. The goal is controlled contrast rather than population representativeness.

Every base also has a neutral synonym replacement. For each cue category, the analysis compares its movement with the control movement for the same base message. This subtracts the background movement produced by changing a word at all.

3.3 Conversation construction

Every pair is rendered in four conversation views. Two primary views place the target after independently worded neutral preludes:

User: I could use your help with a question.
Assistant: Of course. What would you like help with?
User: [target message]

The second shell uses different but semantically similar wording. The other two views place one or two fixed assistant–user exchanges after the target before the readout is taken. In every view, the two variants have the same number and order of messages and differ at exactly one user turn. The full 3B release contains 4,100 rendered conversation rows.

The default readout phrase is used in all four views. Three paraphrases are also evaluated on one current-turn shell, yielding 7,175 pair-by-condition measurements at 3B. The paraphrases vary wording while preserving the requested attribute, for example “Based on this conversation, this user’s age group is” and “My estimate of the user’s age is”. A fixed panel of 505 pairs over 48 bases is evaluated in all three models, keeping the cross-model comparison exactly matched.

3.4 Estimands and inference

For each pair and readout, movement is averaged across five probe seeds. The primary estimate then averages the two current-turn conversation shells. For cue category k and base message i , let M_{ik} be that average and M_{i0} the corresponding random-synonym control. The reported effect is

$$\hat{\tau}_k = \frac{1}{|I_k|} \sum_{i \in I_k} (M_{ik} - M_{i0}).$$

The base message is the independent unit. Confidence intervals use 5,000 base-cluster bootstrap draws. Two-sided p -values use 10,000 sign flips at the base level. Benjamini–Hochberg correction covers the 36 non-control cue-by-attribute cells. Probe seeds and conversation shells are repeated measurements and are not counted as independent observations.

For temporal persistence, we divide the later-turn paired effect by the current-turn paired effect. The ratio is computed as a ratio of means, with a base-cluster bootstrap interval. Conversation-shell and readout-phrase robustness are summarized by Spearman correlation over the full 36-cell matrix and by the mean absolute cell change.

The cross-model endpoints were fixed before either larger cue map was inspected. Each is compared with the 3B panel by (i) Spearman correlation over the 36 primary cells, (ii) sign and

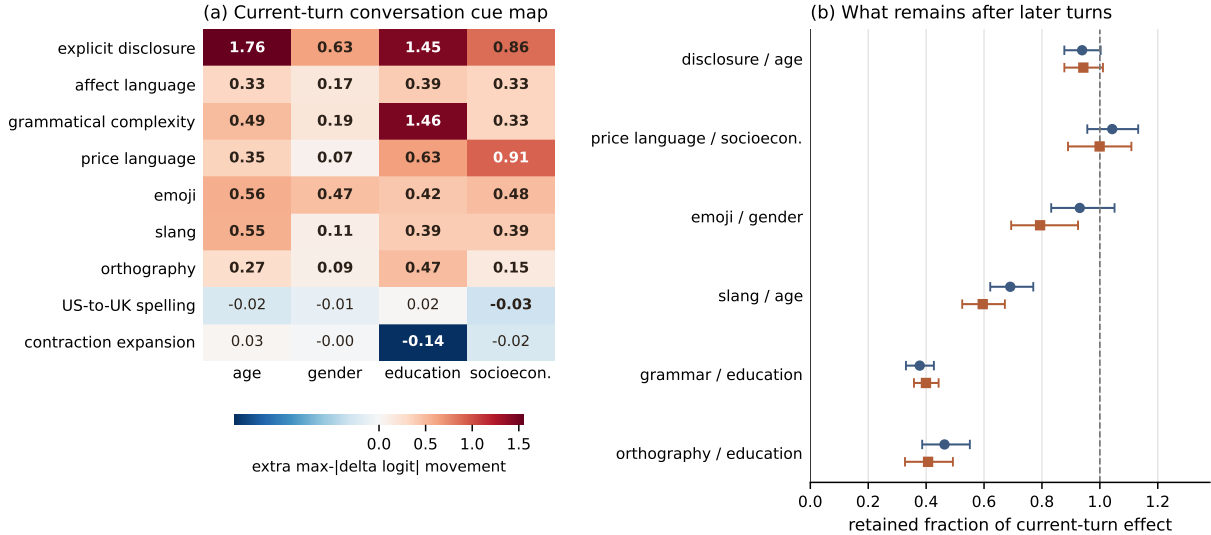


Figure 1: The 1,025-pair 3B experiment. (a) Movement relative to the matched random-edit floor, averaged across two current-turn shells and five full-conversation probe seeds. Bold cells have $q < .05$ after correction over 36 comparisons. (b) Fraction of selected current-turn effects retained after one or two later user turns; bars are base-bootstrap 95% intervals.

corrected-significance agreement for the 30 3B-significant cells, (iii) agreement on the highest-moving readout for each of nine cue categories, and (iv) rank and absolute agreement for the six pre-specified retention cells. We do not attach a cell-independence p -value to the cross-model rank correlation because cells sharing cues and readouts are structured, not independent observations.

4 Results

4.1 The 3B cue map

Figure 1 gives the complete 1,025-pair 3B cue map. Thirty of 36 cells have $q < .05$: 28 move more than their matched synonym controls and two move less. The two negative cells are contraction-expansion-to-education and US-to-UK-spelling-to-socioeconomic-status. Both belong to cue categories whose overall movement is small.

The largest effects align with recognizable kinds of evidence, but the map is not purely diagonal. Disclosure moves age by 1.756 logits above control and education by 1.448. Grammar moves education by 1.458, yet also moves age by 0.490 and socioeconomic status by 0.329. Price language moves socioeconomic status by 0.915 and education by 0.634. Emoji produce broad movement across all four readouts, while slang most strongly moves age.

Two familiar style changes contribute little or less than the lexical control. Contraction expansion ranges from -0.136 to 0.030 across the readouts, and US-to-UK spelling ranges from -0.028 to 0.022 . The model does not treat every visible stylistic difference as equally informative.

Categorical flips depend on the starting margin, but they give an intuitive scale. Grammatical complexity changes the ensemble education label in 27% of its pairs, price language changes socioeconomic status in 37%, emoji change gender in 30%, and slang changes age in 22%.

Cue	Readout	Extra movement	95% CI
Explicit disclosure	age	1.756	[1.515, 1.996]
Grammatical complexity	education	1.458	[1.238, 1.685]
Explicit disclosure	education	1.448	[1.183, 1.734]
Price language	socioeconomic status	0.915	[0.777, 1.056]
Explicit disclosure	socioeconomic status	0.860	[0.726, 1.003]
Price language	education	0.634	[0.475, 0.809]
Emoji	age	0.565	[0.479, 0.659]
Slang	age	0.547	[0.434, 0.672]
Orthography	education	0.474	[0.377, 0.575]

Table 3: Selected primary effects. Extra movement is the category movement minus the matched random-synonym movement, measured in maximum absolute class logit change.

4.2 Cross-model agreement

The cue map has strong cross-model agreement (Figure 2). On the shared panel, the 8B map correlates $\rho = .927$ with 3B; the 32B cross-family map correlates $\rho = .849$. All 30 cells that are corrected-significant at 3B retain a positive above-control effect in both models. Twenty-nine of those 30 remain corrected-significant in each larger model. The 8B miss is contraction-expansion-to-gender; the 32B miss is US-to-UK-spelling-to-education. Across all 36 cells, 32 are corrected-significant at 8B and 33 at 32B.

Model	Rank ρ	Same sign	Still $q < .05$	Same top readout
Llama 3.1 8B	.927	30/30	29/30	8/9
OLMo 2 32B	.849	30/30	29/30	6/9

Table 4: Cross-model endpoints relative to the shared-panel 3B map. Sign and corrected-significance columns cover the 30 3B-significant cells; the last column covers nine cue categories.

The stable core is substantively recognizable. Explicit disclosure most strongly moves age; grammar and orthography most strongly move education; price language most strongly moves socioeconomic status; and slang most strongly moves age in all three models. Contraction expansion also remains age-dominant. The largest categorical exception is emoji: gender is its strongest readout in the 3B panel, while age is strongest at 8B and 32B. In OLMo 2, affect language and US-to-UK spelling are also age-dominant rather than socioeconomic- and education-dominant. The shared structure is therefore strong, but the winning readout for a particular cue is not universal.

4.3 Stability across context and readout wording

The two neutral conversation shells produce almost the same ordering in every model: $\rho = .993$ in the full 3B analysis, .987 at 8B, and .995 at 32B. The result therefore does not depend on one particular prelude.

The readout phrase matters more, though it does not erase the map. Three paraphrases correlate .926–.961 with the default phrase in the full 3B run, .842–.945 at 8B, and .948–.962 at 32B. The rank ordering of cues is robust in this experiment; an individual raw logit magnitude should not be treated as phrase-invariant.

Probe-seed ranges tell a similar story. Large effects keep their direction, but magnitudes vary.

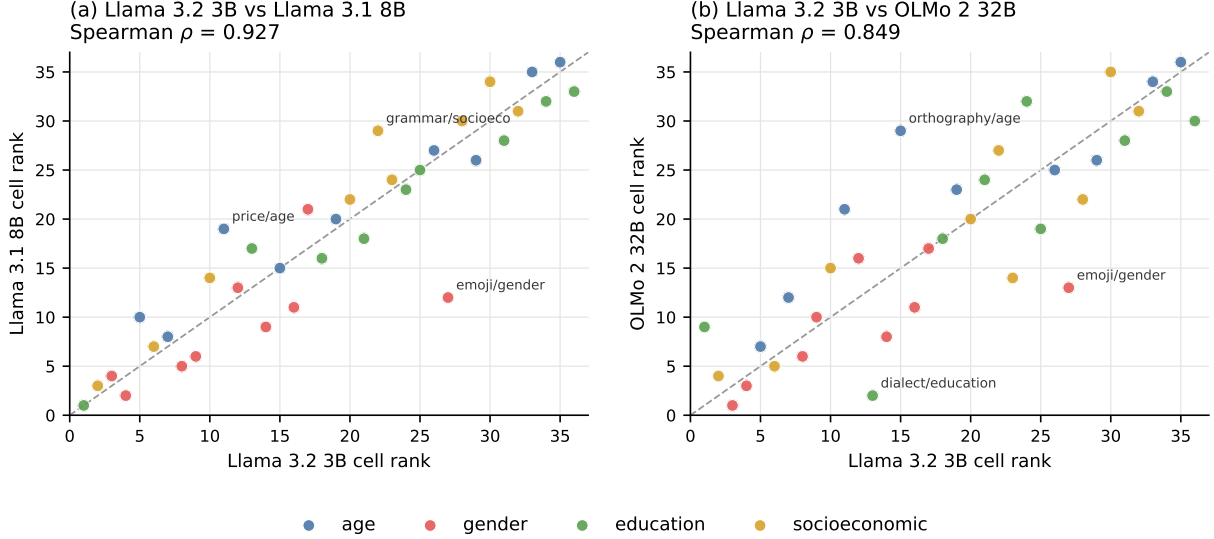


Figure 2: Cross-model agreement on the shared 505-pair, 36-cell cue map. Axes are within-model ranks because independently fitted probe logits do not share a calibrated unit. The dashed line marks identical ranks. Labels identify the three largest rank changes in each comparison.

In the full 3B run, grammar-to-education ranges from 0.447 to 2.474 across seeds, while emoji-to-age ranges from 0.404 to 0.854. The ensemble supports directional and comparative claims more strongly than a universal logit scale.

4.4 What remains after later turns?

The time course separates direct user-relevant content from several surface cues. Table 5 reports the fraction of each current-turn effect retained after later neutral exchanges in all three models.

Disclosure-to-age is highly persistent in every model, retaining .90–.98 of its current-turn effect after two turns. Slang-to-age falls to .47–.57, and orthography-to-education falls to .23–.42. Price-to-socioeconomic-status is nearly unchanged at 3B and 8B but retains .69 at 32B. Grammar-to-education also decays less at 32B than in either Llama model. Across the six pre-specified cells, the 3B retention ordering correlates .886 and .829 with 8B after one and two turns, and .771 at both lags with 32B. The broad distinction between durable content and fading style is visible across models, but its exact time course is model-dependent. The full 3B estimates are similar: after two turns, disclosure-to-age retains .94, price-to-socioeconomic-status 1.00, emoji-to-gender .79, slang-to-age .60, grammar-to-education .40, and orthography-to-education .41.

5 Discussion

The experiment gives a concrete answer to what makes these models revise their readouts of a user. Explicit self-description has the broadest direct effect. Outside disclosure, complex syntax and orthographic care most consistently move education, budget or luxury language moves socioeconomic status, and slang moves age. These assignments appear in the full 3B experiment and under separate probe training in a closely related 8B model and a cross-family 32B model.

Cue / readout	3B	8B	32B
<i>One turn later</i>			
Disclosure / age	1.01	1.04	0.95
Price / socioeconomic status	1.03	1.17	0.71
Emoji / gender	0.97	1.10	0.93
Slang / age	0.68	0.51	0.66
Grammar / education	0.41	0.48	0.64
Orthography / education	0.52	0.44	0.24
<i>Two turns later</i>			
Disclosure / age	0.98	0.98	0.90
Price / socioeconomic status	1.01	1.15	0.69
Emoji / gender	0.79	1.45	0.76
Slang / age	0.57	0.47	0.51
Grammar / education	0.43	0.46	0.66
Orthography / education	0.41	0.42	0.23

Table 5: Ratio of later-turn to current-turn paired effects on the shared 505-pair panel. Values above one indicate greater measured movement at the later readout, not a calibrated gain in memory. Bootstrap intervals for every cell are released with the results.

Several patterns resemble social stereotypes. That is a property of the model and readout, not evidence that a cue is a legitimate basis for judging a person. Emoji illustrate the problem: they are gender-dominant in the 3B shared panel but age-dominant in the full 3B experiment and both larger models. A cue can move several readouts without identifying a demographic fact. Controlled maps expose which harmless choices alter a personalization signal and which apparent rules fail to transfer.

The persistence results add a second dimension. A user impression is not just a classification of the latest turn. Some cues continue to shape the readout after new messages arrive, while others are partially overwritten. Systems that maintain or act on user models should therefore be evaluated not only for which attributes they infer, but for which evidence they retain and how long it remains influential.

The stability checks clarify what is robust. Conversation-shell wording barely changes cue ordering; readout phrasing and probe fitting affect magnitudes. Model changes preserve broad ranks while changing some dominant readouts and retention ratios. The strongest conclusion is relational: under a fixed protocol, some cues reliably move some readouts more than a matched lexical edit. A raw logit unit has no shared psychological meaning across attributes or models.

6 Limitations and ethics

The 100 base messages and their edits are authored. They cover several common assistant-use topics but are not a representative sample of natural dialogue. Fixed assistant turns are necessary to isolate the edited user cue, yet real assistants may respond differently to the two variants and create additional feedback effects.

The three checkpoints do not isolate scale. Two are from the Llama family and one is OLMo 2; size, architecture, pretraining, and post-training all change. The results support cross-model agreement, not a claim that parameter count causes any difference. Proprietary systems and other model families may behave differently.

Minimal pairs isolate a target cue but cannot make every linguistic dimension identical. Grammar rewrites are longer and somewhat more formal; disclosures and price qualifiers intentionally add content; slang can jointly alter era, register, capitalization, and affect. Matched random edits control ordinary lexical movement, not every possible correlated feature.

Linear probes measure decodable structure at a selected hidden state. Their source conversations are synthetic and label-conditioned; both user and assistant language may carry regularities from that generation process. Strong held-out performance does not prove that the labels are portable to every domain or that a model uses the readout when generating an answer. Fixing all assistant messages within each evaluation pair prevents assistant responses from mediating the measured edit, but it does not turn the source task into a natural demographic sample. Suffix-paraphrase variation is a further warning against treating the instrument as context-free.

Age, gender, education, and socioeconomic labels are coarse and normative. Inferring them from style can reproduce stereotypes, erase identities, and create privacy harms (Zaman and Garimella, 2026). The reported outputs must not be presented as justified facts about a person. A practical use of this work is instead to test whether recommendations, moderation, or personalization change under harmless stylistic variation when they should not.

7 Conclusion

Small choices inside a conversation produce reproducible, attribute-specific changes in user readouts across 1,025 controlled pairs and 3B, 8B, and 32B instruction models. Disclosure most strongly moves age, grammar and orthography move education, price language moves socioeconomic status, and slang moves age in the full 3B experiment and on the shared three-model panel. The core map is stable, while emoji and several weaker cues change their dominant readout. Direct disclosure remains readable after later turns; several style effects fade at model-dependent rates.

The result is deliberately narrow: it maps evidence that moves a fixed model impression, not truth about a user. Within that boundary, the breadth of the controlled dataset and the cross-model agreement make the central cue structure more credible while showing why individual cue rules should not be treated as universal.

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